Hw 19

Section 6.1

(1)

(2) $H_0 : \mu \leq 1$ vs $H_1 : \mu > 1$

$n = 50$ $\mu = 1.03$ $\sigma = 0.1$

$$Z = \frac{1.03 - 1}{0.1/\sqrt{50}} = 2.12132$$

P-value $= P(Z > 2.12132) = 0.0170$

(b) Either the mean resistance is greater than 1 mΩ or the sample is in the most extreme 1.7% of its distribution.

(3)

$H_0 : \mu = 232$ vs $H_1 : \mu \neq 232$

(a) $n = 66$, $\bar{x} = 231.7$, $\sigma = 2.19$

$$Z = \frac{231.7 - 232}{2.19/\sqrt{66}} = -1.1122$$

P-value $= 0.2670$

(3) 26.7 %.

(5) $n = 80$, $\mu = 5.4$ $\bar{x} = 4.5$, $\sigma = 2.7$

(a) $H_0: \mu \geq 5.4$ and $H_1: \mu < 5.4$

$$Z = \frac{5.4 - 4.5}{2.7/\sqrt{80}} = -2.985$$

$P = 0.0014$

(b) If the mean number of sick days were 5.4, the prob. is only 0.0014 of observing a sample with a mean less than or equal to the mean of 4.5 that was actually observed. Since this is a very small probability we are convinced that the mean number of sick days is less than 5.4.

(4)

$n = 60$, $\mu = 900$, $\bar{x} = 850$, $\sigma = 153$

(a) $H_0: \mu \geq 900$ vs $H_1: \mu < 900$.

$$Z = \frac{900 - 850}{153/\sqrt{60}} = -2.53$$

P-value $= 0.0057$

(b) If mean depth were 900 μm or more, the prob. of observing a sample mean as small as the value of 850 that was actually observed is 0.0057. Therefore, we are convinced that the mean depth is not 900 μm or more, but instead is less than 900 μm.

Section: 6.2

(1)

P = 0.5 has more plausible.

(3)

(iv) Ho might be true but it's unlikely

(5)

(a) True

(b) False

(c) True

(d) False

(11)

(a) $H_0: \mu \leq 8$ vs $H_1: \mu > 8$

(b) $H_0: \mu \leq 60,000$ vs $H_1: \mu > 60,000$

(c) $H_0: \mu = 10$ vs $H_1: \mu \neq 10$

(12) $n = 75$, $\mu = 4.0$, $\bar{x} = 3.72$, $\sigma = 1.93$

(a)
1. The sample mean is in 2 standard deviation
2. The scale might be out of calibration.

(b) $H_0: \mu \geq 4.0$ vs $H_1: \mu < 4.0$

(c)

$$Z = \frac{3.72 - 4.0}{1.93 / \sqrt{75}} = \frac{-0.28}{0.22285704}$$

$$= 0.125640991$$

P-value = 1.15

(15) No, she cannot conclude that the null hypothesis is true.

(16) $n = 100$ $H_0 : \mu \leq 20$ vs $H_1 : \mu > 20$

$\sigma = 10$. $\bar{x} = 21$

(a) $$Z = \frac{21 - 20}{10/\sqrt{100}} = 1$$

P-value = 0.1587

Because 0.16 is greater than value 0.1, we would fail to reject at all commonly used significance of 0.1, 0.05, 0.01. There is very weak evidence to support the claim.

(b) $n = 1000$

$$Z = \frac{21 - 20}{10/\sqrt{1000}} = 3.16227766$$

P-value = 0.0008

Because 0.0008 is smaller value than 0.001, we would reject at all commonly used significance of 0.1, 0.05, 0.01. There is very strong evidence to reject the claim.